Slide Set #5

Inference based on the CLRM

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Notes

Finite sample conditional inference with the CLRM

$$y_i \;=\; \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots \beta_K x_{ik} + \varepsilon_i \;=\; \boldsymbol{x}_i'\boldsymbol{\beta} + \varepsilon_i$$

for all $i = 1, 2, \ldots, n$, or in matrix form

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Under the CLRM assumptions we investigate how to

- estimate unknown parameters $\boldsymbol{\beta}$ and σ^2
- compute standard errors and confidence intervals for $\boldsymbol{\beta}$
- perform hypothesis tests on $\boldsymbol{\beta}$
- compute and assess predictions for unobserved future values of y
- assess the overall performance of the model

Notes

- $\boldsymbol{y}, \boldsymbol{X}, \varepsilon$ are random variables
- we work with finite samples, $n < +\infty$
- the analysis is always conducted conditioning on $\boldsymbol{X}$

Remark
conditioning on $\boldsymbol{X}$ is a trick that allows us to have only one source of randomness to take care of. In fact, once we condition on $\boldsymbol{X}$, the variations in $\boldsymbol{y}$ are only due to ε

Remark
I do not introduce separate notations for *statistics* and their *estimates*:

- a function $T(\boldsymbol{y}, \boldsymbol{X})$ of observable random quantities is a statistics (that is a random variable)
- when I replace observed values into $T(\cdot)$, I call it the "*observed value of* $T(\cdot)$" rather than having another symbol for it.

Notes

PART I

ESTIMATORS

Notes

OLS estimator for $\boldsymbol{\beta}$

$$S(\boldsymbol{\beta}) = \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}),$$

The OLS estimator $\boldsymbol{b}$ is defined as the optimal solution

$$\boldsymbol{b}(\boldsymbol{X}, \boldsymbol{y}) = \boldsymbol{b} = \underset{\boldsymbol{\beta} \in \mathbb{R}^K}{\arg\min}\, S(\boldsymbol{\beta})$$

The OLS estimator coincides with the solution of the LS-LFP when we consider the random variables instead of the observed values, therefore

$$\boldsymbol{b} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}\boldsymbol{y}$$

Remark
Existence of $\boldsymbol{b}$ is guaranteed by (A3) with probability 1

Remark
The <u>observed value of $\boldsymbol{b}$</u> retains all the geometric/algebraic of the solution to the *linear fitting problem*

Notes

Fitted values, also called the predicted values

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{b}$$

where the components of the vector $\hat{\boldsymbol{y}}$ are

$$\hat{y}_i = b_1 + b_2 x_{i2} + b_2 x_{i2} + \cdots + b_K x_{iK}; \qquad i = 1, 2, \ldots, n$$

These are estimators of the regression function, therefore

$$\hat{y}_i \quad \text{is an estimator for} \quad \mathrm{E}[y_i \,|\, \boldsymbol{X}]$$

Residual

$$\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{b} = \boldsymbol{y} - \hat{\boldsymbol{y}}$$

where

$$e_i = y_i - (b_1 + b_2 x_{i2} + b_2 x_{i2} + \cdots + b_K x_{iK}) \qquad i = 1, 2, \ldots, n$$

⚠ e_i is not an "*estimator*" of the error. But $\{e_1, e_2, \ldots, e_n\}$ can be used to make inference about the error distribution

Notes

Estimator for σ^2 (error variance)

Observe
the sample variance of $\{e_1, e_2, \ldots, e_n\}$ would be

$$\frac{1}{n}\sum_{i=1}^{n}(e_i - \bar{e})^2 = \left(\frac{1}{n}\sum_{i=1}^{n} e_i^2\right) - \left(\frac{1}{n}\sum_{i=1}^{n} e_i\right)^2,$$

however, because of the properties of the LS-LFP solution, the observed sample average of the residuals is always exactly zero (is this consistent with CLRM?)

Unbiased estimator of σ^2

$$s^2 = \frac{1}{n-K}\sum_{i=1}^{n} e_i^2$$

this is a *two-steps* estimator, why?

Notes

PART II

VARIANCE DECOMPOSITION
AND MEASURES OF FIT

Notes

Total variance decomposition

Let $\bar{y}$ be the sample average of the observed $y_1, y_2, \ldots, y_n$:

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i,$$

and let $\bar{\boldsymbol{y}} = (\underbrace{\bar{y}, \bar{y}, \ldots, \bar{y}}_{n \text{ times}})'$

Total sum of squares

$$\text{TSS} = \sum_{i=1}^{n}(y_i - \bar{y})^2 = (\boldsymbol{y} - \bar{\boldsymbol{y}})'(\boldsymbol{y} - \bar{\boldsymbol{y}})$$

This is the deviance (variability) observed in the independent variable y

Observe
TSS does not depend on $\boldsymbol{b}$, it is given by the observed data about $\boldsymbol{y}$ whatever we do with our model

Notes

Residual sum of squares

$$\text{RSS} = \sum_{i=1}^{n} e_i^2$$

where

$$S(\boldsymbol{b}) = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}) = \boldsymbol{e}'\boldsymbol{e}$$

Standard error of the regression

$$\text{SER} = \sqrt{\frac{\text{RSS}}{n-K}} = s$$

Both RSS and SER are measures of *lack of fit*: they measure how close $\hat{\boldsymbol{y}}$ fits $\boldsymbol{y}$. They depend on $\boldsymbol{b}$, therefore they depend on the modeling

Notes

Explained sum of squares

$$\text{ESS} = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 = (\hat{\boldsymbol{y}} - \bar{\boldsymbol{y}})'(\hat{\boldsymbol{y}} - \bar{\boldsymbol{y}})$$

this is the overall deviance of the predicted values of y wrt to the unconditional mean of y.

ESS is a measure of fitting ability of the model. In fact, ESS also depends on the modeling.

Observe
at first look this is not exactly a measure of deviance (why?). But it turns out that another property of LS-LFP is that (see homeworks)

$$\frac{1}{n}\sum_{i=1}^{n} \hat{y}_i = \frac{1}{n}\sum_{i=1}^{n} y_i,$$

therefore it is a measure of the deviance of $\hat{y}_i$

Notes

The R^2 statistic

It can be shown (we don't do this here) that

$$\text{TSS} = \text{ESS} + \text{RSS}$$

From the previous decomposition we get a famous (and misused) goodness of fit statistic

$$\begin{aligned} R^2 &= \frac{\text{ESS}}{\text{TSS}} \\ &= 1 - \frac{\text{RSS}}{\text{TSS}} \end{aligned}$$

R^2 is the portion of deviance observed in the y that is explained by the linear model. This is also called *coefficient of determination*.

Notes

Problems with R^2

- Increases by adding more regressors. For this reason its better to look at the so-called adjusted R^2 (for the degrees of freedom) which is computed as follows:

$$\overline{R}^2 = 1 - \frac{\text{RSS}/(n-K)}{\text{TSS}/(n-1)}$$

- $R^2 \in [0,1]$ only if the constant term is included in the model. So when you estimate without intercept don't be scared if you get $R^2 < 0$
- An extremely large R^2 is pathological, guess why!

Notes

PART III

FINITE SAMPLE PROPERTIES OF $\boldsymbol{b}$ and s^2

Notes

Useful facts

Sampling estimation error for β as function of ε

$$\begin{aligned}
\boldsymbol{b} - \beta &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} - \beta \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'(\boldsymbol{X}\beta + \varepsilon) - \beta \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{X}'\boldsymbol{X})\beta + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\varepsilon - \beta \\
&= \beta + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\varepsilon - \beta \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\varepsilon
\end{aligned}$$

Write $\boldsymbol{e}$ and RSS as a function of ε

$$\boldsymbol{e} = \boldsymbol{M}\boldsymbol{y} = \boldsymbol{M}(\boldsymbol{X}\beta + \varepsilon) = \boldsymbol{M}\varepsilon$$

and

$$\text{RSS} = \boldsymbol{e}'\boldsymbol{e} = (\boldsymbol{M}\varepsilon)'(\boldsymbol{M}\varepsilon) = \varepsilon'\boldsymbol{M}'\boldsymbol{M}\varepsilon = \varepsilon'\boldsymbol{M}\varepsilon$$

Notes

Finite sample properties of $\boldsymbol{b}$

Proposition (unbiasedness of b)

Assume (A1), (A2) and (A3). Then $\mathrm{E}[\boldsymbol{b} \,|\, \boldsymbol{X}] = \beta$, that is $\boldsymbol{b}$ is an unbiased estimator of β conditional on $\boldsymbol{X}$.

Proof. First note that by the linearity of expectations

$$\mathrm{E}[\boldsymbol{b} \,|\, \boldsymbol{X}] = \beta \quad \text{whenever} \quad \mathrm{E}[\boldsymbol{b} - \beta \,|\, \boldsymbol{X}] = \boldsymbol{0}$$

Recall the decomposition of the estimation error

$$\boldsymbol{b} - \beta = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\varepsilon$$

then

$$\mathrm{E}[\boldsymbol{b} - \beta \,|\, \boldsymbol{X}] = \mathrm{E}\left[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\varepsilon \,|\, \boldsymbol{X}\right]$$

pull out what's known and obtain

$$\mathrm{E}[\boldsymbol{b} - \beta \,|\, \boldsymbol{X}] = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\,\mathrm{E}\left[\varepsilon \,|\, \boldsymbol{X}\right] = \boldsymbol{0}$$

■

Notes

Observe
this is a conditional unbiasedness statement, and it is stronger than the more traditional unconditional statement.

This is stronger because

$$\mathrm{E}[\boldsymbol{b} \,|\, \boldsymbol{X}] = \boldsymbol{\beta} \implies \mathrm{E}[\mathrm{E}[\boldsymbol{b} \,|\, \boldsymbol{X}]] = \mathrm{E}[\boldsymbol{b}] = \boldsymbol{\beta}$$

Notes

Proposition (variance matrix of b)
Under (A1), (A2), (A3), and (A4), $\mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}] = \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$

Proof. Since $\boldsymbol{\beta}$ is not a random variable, then its variance is zero, then

$$\begin{aligned}
\mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}] &= \mathrm{Var}[\boldsymbol{b} - \boldsymbol{\beta} \,|\, \boldsymbol{X}] \\
&= \mathrm{Var}[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}] \\
&= \mathrm{Var}[\boldsymbol{A}\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}], \quad \text{for } \boldsymbol{A} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \\
&= \boldsymbol{A}\mathrm{Var}[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}]\boldsymbol{A}' \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \,\mathrm{Var}[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}]\, \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} \qquad (*) \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \,(\sigma^2)\boldsymbol{I}_n\, \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} \\
&= \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\,(\boldsymbol{X}'\boldsymbol{X})\,(\boldsymbol{X}'\boldsymbol{X})^{-1} \\
&= \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}
\end{aligned}$$

(*) because $(\boldsymbol{B}')^{-1} = (\boldsymbol{B}^{-1})'$ ■

Notes

Proposition (Gauss-Markov theorem)

Assume (A1), (A2), (A3), and (A4). Let $\boldsymbol{b}_0$ be any estimator of $\boldsymbol{\beta}$ that is linear in the $\boldsymbol{y}$ and unbiased. Then

$$\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] \succeq \mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$$

Remark
Since $\boldsymbol{b}$ is unbiased, its MSE (efficiency) is equal to its variance $\mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$

Recall
$\succeq$ means bigger or equal for certain matrices. Since of **(b)**

Gauss-Markov Theorem says that OLS are BLUE
in terms of MSE the OLS is the *Best Linear Unbiased Estimator* (BLUE)

Notes

Proof. First note that OLS is linear map of $\boldsymbol{y}$. In fact, define $\boldsymbol{A} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$, then $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{y}$.

By assumption there exists an alternative linear estimator $\boldsymbol{b}_0$, wlog assume it is $\boldsymbol{b}_0 = \boldsymbol{C}\boldsymbol{y}$ for some matrix $\boldsymbol{C}$.

For any $\boldsymbol{C}$ we can always find a matrix $\boldsymbol{D} = \boldsymbol{C} - \boldsymbol{A}$ so that

$$\boldsymbol{b}_0 \;=\; \boldsymbol{C}\boldsymbol{y} \;=\; (\boldsymbol{D} + \boldsymbol{A})\boldsymbol{y}$$

By (A1) $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, therefore

$$\begin{aligned}\boldsymbol{b}_0 =& (\boldsymbol{D} + \boldsymbol{A})\boldsymbol{y} \\ =& \boldsymbol{D}\boldsymbol{y} + \boldsymbol{A}\boldsymbol{y} \\ =& \boldsymbol{D}(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) + \boldsymbol{b} \\ =& \boldsymbol{D}\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{D}\boldsymbol{\varepsilon} + \boldsymbol{b}\end{aligned}$$

By assumption $\boldsymbol{b}_0$, it is unbiased conditional on $\boldsymbol{X}$, i.e. $\mathrm{E}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] = \boldsymbol{\beta}$.

Notes

Therefore

$$\begin{aligned}\mathrm{E}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] &= \mathrm{E}[\, \boldsymbol{D}\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{D}\boldsymbol{\varepsilon} + \boldsymbol{b} \,\,|\, \boldsymbol{X}] \\ &= \boldsymbol{D}\boldsymbol{X}\boldsymbol{\beta} + \mathrm{E}[\boldsymbol{D}\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}] + \mathrm{E}[\boldsymbol{b} \,|\, \boldsymbol{X}] \\ &= \boldsymbol{D}\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{D}\,\mathrm{E}[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}] + \boldsymbol{\beta} \\ &= \boldsymbol{D}\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\beta}\end{aligned}$$

The fact that $\boldsymbol{b}_0$ is unbiased implies $\boldsymbol{D}\boldsymbol{X} = \boldsymbol{0}$, hence

$$\boldsymbol{b}_0 = \boldsymbol{D}\boldsymbol{\varepsilon} + \boldsymbol{b} \tag{5.1}$$

Now substract $\boldsymbol{\beta}$ from both side of (5.1) and obtain the estimation error $\boldsymbol{b}_0$

$$\boldsymbol{b}_0 - \boldsymbol{\beta} \;=\; \boldsymbol{D}\boldsymbol{\varepsilon} + (\boldsymbol{b} - \boldsymbol{\beta}) \;=\; (\boldsymbol{D} + \boldsymbol{A})\boldsymbol{\varepsilon}$$

Notes

In order to show that the OLS is BLUE we need to show that for any $\boldsymbol{b}_0$ (linear in $\boldsymbol{y}$ and unbiased) it holds true that $\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}]$ is bigger then $\mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$

We need to show that $\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] \succeq \mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$, that is

$$\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] - \mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}] \quad \text{is PSD.}$$

Same trick as before: the variance of a constant is zero, so we start from $\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] = \mathrm{Var}[(\boldsymbol{b}_0 - \boldsymbol{\beta}) \,|\, \boldsymbol{X}]$.

Notes

$$
\begin{aligned}
\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] &= \mathrm{Var}[(\boldsymbol{b}_0 - \boldsymbol{\beta}) \,|\, \boldsymbol{X}] \\
&= \mathrm{Var}[(\boldsymbol{D} + \boldsymbol{A})\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}] \\
&= (\boldsymbol{D} + \boldsymbol{A})\mathrm{Var}[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}](\boldsymbol{D} + \boldsymbol{A})' \\
&= \sigma^2(\boldsymbol{D} + \boldsymbol{A})(\boldsymbol{D}' + \boldsymbol{A}') \\
&= \sigma^2(\boldsymbol{D}\boldsymbol{D}' + \boldsymbol{A}\boldsymbol{D}' + \boldsymbol{D}\boldsymbol{A}' + \boldsymbol{A}\boldsymbol{A}')
\end{aligned}
$$

Now

$$\boldsymbol{D}\boldsymbol{A}' = \boldsymbol{D}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} = \boldsymbol{0}$$

$$\boldsymbol{A}\boldsymbol{A}' = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} = (\boldsymbol{X}'\boldsymbol{X})^{-1}$$

$$\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] = \sigma^2[\boldsymbol{D}\boldsymbol{D}' + (\boldsymbol{X}'\boldsymbol{X})^{-1}] = \sigma^2\boldsymbol{D}\boldsymbol{D}' + \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$$

Notes

Observe that

- any covariance matrix must be PSD
- the sum of PSD matrices is PSD

Since

$$\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] = \sigma^2\boldsymbol{D}\boldsymbol{D}' + \mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$$

$\sigma^2\boldsymbol{D}\boldsymbol{D}'$ must be PSD

But

$$\sigma^2\boldsymbol{D}\boldsymbol{D}' = \mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] - \mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$$

which implies that the right-hand-side must be PSD. Hence

$$\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] \succeq \mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$$

which completes the proof ■

Notes

Proposition (b is uncorrelated with e)

Assume (A1), (A2) and (A3) and (A4), then $\mathrm{Cov}[\boldsymbol{b}, \boldsymbol{e} \mid \boldsymbol{X}] = \boldsymbol{0}$

Proof. Recall that $\boldsymbol{e} = \boldsymbol{M}\boldsymbol{\varepsilon}$,

Now work out the formula of covariance between vectors:

$$
\begin{aligned}
\mathrm{Cov}[\boldsymbol{b}, \boldsymbol{e} \mid \boldsymbol{X}] &= \mathrm{E}[(\boldsymbol{b} - \mathrm{E}[\boldsymbol{b}|\boldsymbol{X}])(\boldsymbol{e} - \mathrm{E}[\boldsymbol{e}|\boldsymbol{X}])' \mid \boldsymbol{X}] \\
&= \mathrm{E}[(\boldsymbol{b} - \boldsymbol{\beta})(\boldsymbol{e})' \mid \boldsymbol{X}] \\
&= \mathrm{E}[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon}\,(\boldsymbol{M}\boldsymbol{\varepsilon})' \mid \boldsymbol{X}] \\
&= \mathrm{E}[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon}\,\boldsymbol{\varepsilon}'\boldsymbol{M} \mid \boldsymbol{X}] \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\,\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \boldsymbol{X}]\boldsymbol{M} \quad \text{(pull out what's known)} \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\sigma^2\boldsymbol{I}_n\boldsymbol{M} \\
&= \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{M}
\end{aligned}
$$

Since $\boldsymbol{X}'\boldsymbol{M} = (\boldsymbol{M}\boldsymbol{X})' = \boldsymbol{0}$, then $\mathrm{Cov}[\boldsymbol{b}, \boldsymbol{e} \mid \boldsymbol{X}] = \boldsymbol{0}$ ■

Notes

Finite sample properties of s^2

Proposition (s^2 is unbiased)

Assume (A1), (A2), (A3), (A4), then s^2 is an unbiased estimator of σ^2.

Proof.

$$
\begin{aligned}
\mathrm{E}[\boldsymbol{e}'\boldsymbol{e} \mid \boldsymbol{X}] &= \mathrm{E}[(\boldsymbol{M}\boldsymbol{\varepsilon})'\boldsymbol{M}\boldsymbol{\varepsilon} \mid \boldsymbol{X}] \\
&= \mathrm{E}[\boldsymbol{\varepsilon}'\boldsymbol{M}'\boldsymbol{M}\boldsymbol{\varepsilon} \mid \boldsymbol{X}] \\
&= \mathrm{E}[\boldsymbol{\varepsilon}'\boldsymbol{M}\boldsymbol{\varepsilon} \mid \boldsymbol{X}] \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} m_{ij}\,\mathrm{E}[\varepsilon_i\varepsilon_j \mid \boldsymbol{X}] \quad \text{(verify this!)} \\
&= \sigma^2\sum_{i=1}^{n} m_{ii} \\
&= \sigma^2\,\mathrm{tr}(\boldsymbol{M})
\end{aligned}
$$

Notes

$$\mathrm{tr}(\boldsymbol{M}) = \mathrm{tr}(\boldsymbol{I}_n - \boldsymbol{P}) = n - \mathrm{tr}(\boldsymbol{P})$$

and

$$\mathrm{tr}(\boldsymbol{P}) = \mathrm{tr}\left(\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\right) = \mathrm{tr}\left(\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\right) = \mathrm{tr}(\boldsymbol{I}_k) = K$$

Therefore $\mathrm{E}[\boldsymbol{e}'\boldsymbol{e} \,|\, \boldsymbol{X}] = (n-K)\sigma^2$. And this proves that the analog estimator $\hat{\sigma}^2$ is biased, in fact

$$\mathrm{E}\left[\hat{\sigma}^2 \,|\, \boldsymbol{X}\right] = \mathrm{E}\left[\frac{\boldsymbol{e}'\boldsymbol{e}}{n} \,\middle|\, \boldsymbol{X}\right] = \frac{n-K}{n}\sigma^2$$

This also shows that s^2 is unbiased, in fact

$$\mathrm{E}[s^2 \,|\, \boldsymbol{X}] = \mathrm{E}\left[\frac{\boldsymbol{e}'\boldsymbol{e}}{n-K} \,\middle|\, \boldsymbol{X}\right] = \frac{n-K}{n-K}\sigma^2 = \sigma^2$$

■

Notes

Estimating the standard errors of $\boldsymbol{b}$

We know that

$$\mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}] = \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$$

which depends on population quantities.

We estimate the variance matrix of $\boldsymbol{b}$ based on the plug-in principle:

$$\widehat{\mathrm{Var}\,(\boldsymbol{b})} = s^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$$

Standard error of b_k
Let $(\boldsymbol{X}'\boldsymbol{X})^{-1}_{kk}$ the kth element on the diagonal of $(\boldsymbol{X}'\boldsymbol{X})^{-1}$.

$$\mathrm{Se}(b_k) = s\sqrt{(\boldsymbol{X}'\boldsymbol{X})^{-1}_{kk}}$$

Notes

PART IV

CONFIDENCE INTERVALS
AND HYPOTHESIS TESTING

Notes

Distributional assumption

Inference in finite sample is not distribution-free, we need to assume something which leads to well defined inference

Assumption (A5): Gaussian errors

$\varepsilon \mid \boldsymbol{X} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$

Remarks:

- (A5) is consistent with previous (A2), (A4)
- the distribution of $\varepsilon \mid \boldsymbol{X}$ does not depend on $\boldsymbol{X}$

Notes

Proposition (distribution of b)

Assume (A1)–(A5)

$$(\boldsymbol{b}-\boldsymbol{\beta})\,|\,\boldsymbol{X} \sim \mathcal{N}\left(\boldsymbol{0}, \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\right)$$

Proof. From the estimation error decomposition

$$(\boldsymbol{b}-\boldsymbol{\beta}) = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon} = \boldsymbol{A}\boldsymbol{\varepsilon}, \qquad \text{where} \quad \boldsymbol{A} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'.$$

By (A5) plus the linearity property of the multivariate normal, $(\boldsymbol{b}-\boldsymbol{\beta})\,|\,\boldsymbol{X}$ is normally distributed with parameters

$$\mathrm{E}[(\boldsymbol{b}-\boldsymbol{\beta})\,|\,\boldsymbol{X}] = \mathrm{E}[\boldsymbol{A}\boldsymbol{\varepsilon}\,|\,\boldsymbol{X}] = \boldsymbol{A}\,\mathrm{E}[\boldsymbol{\varepsilon}\,|\,\boldsymbol{X}] = \boldsymbol{0}$$

$$\mathrm{Var}[(\boldsymbol{b}-\boldsymbol{\beta})\,|\,\boldsymbol{X}] \;=\; \mathrm{Var}[\boldsymbol{A}\boldsymbol{\varepsilon}\,|\,\boldsymbol{X}] \;=\; \boldsymbol{A}\,\mathrm{Var}[\boldsymbol{\varepsilon}\,|\,\boldsymbol{X}]\boldsymbol{A}'$$

$$\begin{aligned}\boldsymbol{A}\,\mathrm{Var}[\boldsymbol{\varepsilon}\,|\,\boldsymbol{X}]\boldsymbol{A}' =& (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'(\sigma^2\boldsymbol{I})((\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}')' \\ =& \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} \\ =& \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\end{aligned}$$

which completes the proof ■

Notes

Important property of the multivariate normal distribution: all its marginals are also normally distributed.

Therefore, all the coordinates of the vector $(b_k - \beta_k)$ have a univariate normal distribution

$$(b_k-\beta_k)\,|\,\boldsymbol{X} \sim \mathcal{N}\left(0, \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}_{kk}\right)$$

These distributional results can be used to make inference on $\boldsymbol{\beta}$

Notes

General remark on testing

Classical hypothesis testing (aka Neyman–Pearson optimality):

- Fix $\alpha = \Pr\{\text{false rejection}\} = \Pr\{\text{Type-1 error}\}$
- Max Power $=$ Max $\Pr\{\text{correct rejection}\} =$ Max $\Pr\{\overline{\text{Type-2 error}}\}$

Calculations of $\Pr\{\cdot\}$ requires knowledge of probability laws for underlying data generating process.

We always test more than just H_0-vs-H_1, because in reality we calculate

- $\alpha = \Pr\{\text{reject } \mathsf{H}_0 \,|\, (\mathsf{H}_0 \text{ is true}) \;\&\; (\text{fullfiled model assumptions})\}$
- Power$=\Pr\{\text{reject } \mathsf{H}_0 \,|\, (\mathsf{H}_1 \text{ is true}) \;\&\; (\text{fulfilled model assumptions})\}$

The following testing decision rules are optimal in the sense of Neyman–Pearson only if the underlying data generating process fulfills all/some of A1–A5.

Notes

Testing individual regression coefficients

Tests for individual parameters

$$\mathsf{H}_0 : \beta_k = \beta_k^*$$

t-ratio or *t-test* statistic

$$t_k = \frac{b_k - \beta_k^*}{\sqrt{s^2(\boldsymbol{X}'\boldsymbol{X})^{-1}_{kk}}} = \frac{b_k - \beta_k^*}{\mathrm{Se}(b_k)}$$

It can be shown that

$$t_k \,|\, \boldsymbol{X} \overset{\mathsf{H}_0}{\sim} \mathsf{T}(n-K)$$

Notes

Default test

$$\begin{cases} \mathsf{H}_0 : \beta_k = 0 \\ \mathsf{H}_1 : \beta_k \neq 0 \end{cases}$$

Decision rule at significance level α:

$$\text{reject } \mathsf{H}_0 \text{ if } |t_k| > q_{1-\frac{\alpha}{2}}$$

where $q_{1-\frac{\alpha}{2}}$ is the $(1-\frac{\alpha}{2})$-quantile of the $\mathsf{T}(n-K)$

Confidence interval

Let $(1-\alpha)$ be the confidence level. The confidence interval for β_k is given by

$$b_k \ \pm \ q_{1-\frac{\alpha}{2}} \, \mathrm{Se}(b_k)$$

Notes

Testing linear hypothesis

Many tests of interest can be written in terms of a linear system. Suppose we have the linear model

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \varepsilon$$

We want to test the joint hypothesis that $\beta_1 = \beta_2 = \beta_3 = 0$. This can be formulated as a system of 3 equations: $\{\beta_1 = 0, \beta_2 = 0, \beta_3 = 0\}$ For an appropriate choice of a (3×5)-matrix $\boldsymbol{R}$, and a (3×1)-column vector $\boldsymbol{r}$, we write the null hypothesis in matrix form as

$$\boldsymbol{R}\boldsymbol{\beta} = \boldsymbol{r} \quad \text{that is} \quad \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{that is} \begin{cases} \beta_1 = 0 \\ \beta_2 = 0 \\ \beta_3 = 0 \end{cases}$$

Notes

We consider a set of J linear restrictions of the form

$$\mathsf{H}_0 : \boldsymbol{R}\boldsymbol{\beta} = \boldsymbol{r}$$

where $\boldsymbol{R} = (R_{ij})$ is a $(J \times K)$ matrix, and $\boldsymbol{r}$ is a $(J \times 1)$ column vector

That is, under the null hypotesis is true that

$$\begin{cases} R_{11}\beta_1 + R_{12}\beta_2 + \cdots + R_{1K}\beta_K = & r_1 \\ R_{21}\beta_1 + R_{22}\beta_2 + \cdots + R_{2K}\beta_K = & r_2 \\ \vdots & \\ R_{J1}\beta_1 + R_{J2}\beta_2 + \cdots + R_{JK}\beta_K = & r_J \end{cases}$$

Assume $J < K$, and that $\boldsymbol{R}$ is full row-rank, this means that *rows of* $\boldsymbol{R}$ *are linearly independent*.

Notes

Strategies:

Wald-type testing approach

- estimate the unrestricted model
- based on unrestricted estimates construct a test-statistic that measures how close to the restricted case we are

LRT-type testing approach

- estimate the unrestricted model
- estimate the restricted model
- construct a test-statistic that measures the loss of fit

Notes

Wald's type testing

H_0 is likely to hold when $(\boldsymbol{Rb} - \boldsymbol{r})$ is small enough, or close to $\mathbf{0}$. Hence, under H_0 we expect the norm $\|\boldsymbol{Rb} - \boldsymbol{r}\|$ to be small

The components of $\boldsymbol{b}$ are usually highly correlated, so it does $\boldsymbol{Rb}$. Therefore, the Euclidean norm is not appropriate.

We need an *orthogonalization* device, namely

$$(\boldsymbol{Rb} - \boldsymbol{r})'\boldsymbol{Q}^{-1}(\boldsymbol{Rb} - \boldsymbol{r})$$

that is the Mahalanobis distance of $\boldsymbol{Rb}$ from $\boldsymbol{r}$ for an appropriate scaling matrix $\boldsymbol{Q} = \mathrm{Var}[\boldsymbol{Rb}]$

This kind of testing strategy is called *Wald-test*. It is much more general than this particular case

Notes

The popular F-statistic is a Wald-statistics

Consider the *F-ratio*

$$F = \frac{(\boldsymbol{Rb} - \boldsymbol{r})'\Big[\boldsymbol{R}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}'\Big]^{-1}(\boldsymbol{Rb} - \boldsymbol{r}) \ / \ J}{s^2}$$

Let $\boldsymbol{Q}^{-1} = \Big[s^2\boldsymbol{R}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}'\Big]^{-1}$, then

$$F = \frac{1}{J}(\boldsymbol{Rb} - \boldsymbol{r})'\boldsymbol{Q}^{-1}(\boldsymbol{Rb} - \boldsymbol{r})$$

We will show that under H_0, $\boldsymbol{Q}$ is in fact an estimate of $\mathrm{Var}[\boldsymbol{Rb} - \boldsymbol{r}]$

Notes

Proposition (distribution of the F-statistic)

Assume (A1)–(A5) and H_0, then $F \,|\, \boldsymbol{X} \overset{\mathsf{H}_0}{\sim} \mathsf{F}(J, n-K)$

Proof. Under H_0 $\boldsymbol{r} = \boldsymbol{R}\boldsymbol{\beta}$, therefore

$$\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r} = \boldsymbol{R}\boldsymbol{b} - \boldsymbol{R}\boldsymbol{\beta} = \boldsymbol{R}(\boldsymbol{b} - \boldsymbol{\beta})$$

Moreover since $(\boldsymbol{b} - \boldsymbol{\beta})$ is normally distributed

$$(\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r}) \,|\, \boldsymbol{X} \overset{\mathsf{H}_0}{\sim} \boldsymbol{R}\, \mathcal{N}\left(\boldsymbol{0}, \sigma^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}\right)$$

Which becomes

$$(\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r}) \,|\, \boldsymbol{X} \sim \mathcal{N}\left(\boldsymbol{0}, \sigma^2 \boldsymbol{R}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}'\right)$$

Then

$$(\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r})' \left[\sigma^2 \boldsymbol{R}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}'\right]^{-1} (\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r}) \sim \chi^2(J)$$

Notes

By A5

$$\frac{\boldsymbol{\varepsilon}}{\sigma} \,\Big|\, \boldsymbol{X} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_n)$$

Consider the residual maker matrix $\boldsymbol{M}$, since $\operatorname{rank}(\boldsymbol{M}) = n - K$

$$\left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right)' \boldsymbol{M} \left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right) \,\Big|\, \boldsymbol{X} = \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_n)' \boldsymbol{M}\, \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_n) \sim \chi^2(n-K)$$

Now recall that we can write $\boldsymbol{e}'\boldsymbol{e} = \boldsymbol{\varepsilon}'\boldsymbol{M}\boldsymbol{\varepsilon}$, therefore

$$\frac{\boldsymbol{e}'\boldsymbol{e}}{\sigma^2} \sim \chi^2(n-K)$$

And

$$\frac{s^2}{\sigma^2} = \frac{1}{\sigma^2}\frac{\boldsymbol{e}'\boldsymbol{e}}{n-k} \sim \frac{\chi^2(n-K)}{n-K}$$

Rewrite F as

$$F = \frac{(\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r})' \left[\sigma^2 \boldsymbol{R}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}'\right]^{-1} (\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r}) \,/\, J}{\frac{s^2}{\sigma^2}}$$

Notes

Multiply and divide F by $\frac{1}{\sigma^2}$

$$F = \frac{(\boldsymbol{Rb} - \boldsymbol{r})' \left[\sigma^2 \boldsymbol{R}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}'\right]^{-1} (\boldsymbol{Rb} - \boldsymbol{r}) \;/\; J}{\frac{s^2}{\sigma^2}}$$

Then

$$F \,|\, \boldsymbol{X} = \frac{\frac{\chi^2(J)}{J}}{\frac{\chi^2(n-K)}{n-K}} \overset{\mathsf{H}_0}{\sim} \mathsf{F}(J, n-K)$$

because numerator and denominator are independent given $\boldsymbol{X}$, in fact their conditional distributions do not depend on $\boldsymbol{X}$ ■

Notes

Test

$$\begin{cases} \mathsf{H}_0 : \boldsymbol{R}\beta = \boldsymbol{r} \\ \mathsf{H}_1 : \boldsymbol{R}\beta \neq \boldsymbol{r} \end{cases}$$

Decision rule at significance level α:

$$\text{reject } \mathsf{H}_0 \text{ if } f > q_{1-\alpha}$$

where

- f is the observed value of the F-statitic
- $q_{1-\alpha}$ is the $(1-\alpha)$-quantile of the $\mathsf{F}(J, n-K)$ distribution

Notes

PART V

PREDICTION

Notes

Suppose we want to predict Y knowing that the regressor vector X will be some value $\boldsymbol{x}^0$.

The optimal predictor theorem (OPT) tells us that the best prediction (in the MSE sense) is $\mathrm{E}[Y \mid X = \boldsymbol{x}^0]$, an estimate for this predictor is given by the estimated regression function at $\boldsymbol{x}^0$

$$\widehat{\mathrm{E}[Y \mid X = \boldsymbol{x}^0]} = \hat{y}^0 = \boldsymbol{b}'\boldsymbol{x}^0$$

The OPT says that this is the minimum MSE predictor. Since it is unbiased (why?) $\implies$ it's the minimum variance predictor!

The prediction error is given by $\hat{y}^0 - y^0$ where

$$y^0 = \boldsymbol{\beta}'\boldsymbol{x}^0 + \varepsilon^0 \qquad \text{and} \qquad \hat{y}^0 = \boldsymbol{b}'\boldsymbol{x}^0$$

Therefore

$$e^0 = \hat{y}^0 - y^0 = (\boldsymbol{b} - \boldsymbol{\beta})'\boldsymbol{x}^0 - \varepsilon^0$$

Notes

Proposition (bias/variance of the prediction error)

Assume (A1)–(A4). Let e^0 be the prediction error of the the optimal linear predictor at a point $\boldsymbol{x}^0$, then

1. $\mathrm{E}[e^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] = 0$
2. $\mathrm{Var}[e^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] = \boldsymbol{x}^{0\prime}\left(\sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\right)\boldsymbol{x}^0 + \sigma^2$

Proof.

$$\begin{aligned}\mathrm{E}[e^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] &= \mathrm{E}[(\boldsymbol{b}-\boldsymbol{\beta})'\boldsymbol{x}^0 - \varepsilon^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] \\ &= \mathrm{E}[(\boldsymbol{b}-\boldsymbol{\beta})' \mid \boldsymbol{X}, \boldsymbol{x}^0]\boldsymbol{x}^0 - \mathrm{E}[\varepsilon^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] \\ &= 0\end{aligned}$$

$$\begin{aligned}\mathrm{Var}[e^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] &= \mathrm{Var}[(\boldsymbol{b}-\boldsymbol{\beta})'\boldsymbol{x}^0 - \varepsilon^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] \\ &= \mathrm{Var}[(\boldsymbol{b}-\boldsymbol{\beta})'\boldsymbol{x}^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] + \mathrm{Var}[\varepsilon^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] \\ &= \mathrm{Var}[(\boldsymbol{b}-\boldsymbol{\beta})'\boldsymbol{x}^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] + \sigma^2 \\ &= \boldsymbol{x}^{0\prime}\left(\sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\right)\boldsymbol{x}^0 + \sigma^2\end{aligned}$$

Notes

$\mathrm{Var}[e^0 \mid \boldsymbol{X}, \boldsymbol{x}^0]$ can be estimated with s^2 replacing σ^2.

$\mathrm{Se}(e^0) = \sqrt{\mathrm{Var}[e^0 \mid \boldsymbol{X}, \boldsymbol{x}^0]}$ is the standard error of the prediction error

Prediction confidence interval

Consider the prediction at $\boldsymbol{x}^0$. The $(1-\alpha)$ confidence interval for the predicted value is

$$\hat{y}^0 \pm q_{(1-\frac{\alpha}{2})}\,\mathrm{Se}(e^0)$$

where $q_{1-\frac{\alpha}{2}}$ is the $(1-\frac{\alpha}{2})$-quantile of the $\mathsf{T}(n-K)$ distribution.

Notes